

# UKAIC AI Foundation Student Handbook

A complete revision companion for the eight-domain AI Foundation course.

Created in conjunction with **My Cyber Academy (MCA)**, whose mascot and mark appear on this cover.

Eight domains · 43 sub-areas · 120-question final examination

Every figure in this handbook is generated from the same narration scripts as the course videos, so the two always agree.



# Contents

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| D1     | Understanding AI: The Big Picture         | 12%    | 14             |
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| D3     | The AI Landscape & Generative AI          | 12%    | 14             |
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| D5     | Working with AI Day to Day                | 12%    | 15             |
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| D8     | Ethics, Bias, Responsible AI & Governance | 8%     | 10             |

Total delivery: 164 minutes of video across eight domains. The final examination is 120 multiple-choice questions, 120 minutes, closed book, with a pass mark of 70 per cent.

## Domain 1

# Understanding AI: The Big Picture

Weight 12% · 14 exam questions · about 20 minutes

Accurate, vendor-neutral vocabulary and framing for artificial intelligence — what AI is, what it is not, and how the terms nest.

**AI is not one invention. It is a family of techniques that find patterns in data.**

## 1. What AI actually is

Sub-area 1.1 — defining artificial intelligence.

**A machine doing tasks we associate with human intelligence — by learning from examples, not by following written rules.**

### The filter in your inbox

Nobody writes the rule for tomorrow's scam. The system learns from examples, then predicts. That is machine learning at work, and it has been quietly protecting your inbox for years.

**Asked:** Here are a thousand messages flagged as spam, and a thousand that are perfectly genuine

**Answer:** It works the pattern out for itself — and catches the next dodgy email nobody has ever seen

### Two ways to build software

|   | Written rules                            | Learned patterns                                  |
|---|------------------------------------------|---------------------------------------------------|
| 1 | A person decides the logic up front      | The system is shown examples and infers the logic |
| 2 | It behaves identically every single time | Its behaviour depends on what it learned          |
| 3 | It breaks on anything unplanned          | It fails when its examples misled it              |
| 4 | Example: a spreadsheet formula           | Example: the spam filter                          |

**Guess. Compare with the truth. Adjust. Repeat — millions of times.**

### The recommendation engine

Streaming and video platforms have built their whole business on this. It is narrow AI: one job, done extraordinarily well, at enormous scale.

**Asked:** Watch what this person chooses to watch, and what they skip

**Answer:** Predict the next thing they'll want — a guess made thousands of times a second, for millions of people

### The common trap

A system can match patterns beautifully and understand nothing at all. Fluent output is not comprehension.

**Not conscious. Not infallible. Not magic. And not about to replace your judgement.**

## 2. The four nested terms

Sub-area 1.2 — AI, machine learning, deep learning and generative AI.

## One family, four sizes

- **Artificial intelligence** — the whole field
- **Machine learning** — improves from data
- **Deep learning** — uses layered networks
- **Generative AI** — produces new content

**Software that gets better at a task by being shown examples, instead of being handed rules.**

## AlphaFold — a genuine breakthrough

AlphaFold is still narrow AI. It does one thing: predict protein shapes. It cannot write a shopping list. But at that one job, it changed an entire field of science.

**Asked:** Predict the three-dimensional shape of a protein from its chain of amino acids

**Answer:** In 2020, DeepMind's AlphaFold did it — solving a problem biologists had chased for fifty years

**Machine learning with layers — simple units stacked up, each passing what it found to the next.**

## Translation that suddenly worked

For decades, translation was programmed rule by rule, and it was clunky. Once deep learning was applied, it improved dramatically. Same task, entirely different method.

**Asked:** Translate a sentence from one language into another, and do it well

**Answer:** Around 2016, translation quality jumped — because the system learned from millions of real sentence pairs instead of grammar rules

**Deep learning that produces new content — a paragraph, a picture, a piece of audio, or code.**

## The tool that drafts your words

It is producing text, not retrieving it. Nothing is being looked up. The words are generated, one at a time, from patterns. That's why it can sound right and still be wrong.

**Asked:** Write a polite email asking for an extension on a deadline

**Answer:** A fluent draft appears in seconds — not because it knows you, but because it has seen millions of emails

## Precision

Does it learn from data? Does it use layered networks? Does it make new content? And what single task is it actually for?

**Saying “AI” when you mean “chatbots” is the most common slip in the room.**

## 3. Narrow AI and general AI

Sub-area 1.3 — narrow versus general intelligence.

**Built for one job, or a narrow family of jobs — and every system in service today is narrow.**

## Superhuman at exactly one thing

Go has more possible positions than there are atoms in the universe. Both machines were extraordinary at one game, and helpless at everything else. That gap is what 'narrow' means.

**Asked:** Play chess. Then play Go. Against the best players alive

**Answer:** Deep Blue beat the world chess champion in 1997; AlphaGo beat the Go champion in 2016 — neither can do anything else

## Narrow against general

|   | Narrow AI                                     | General AI                                |
|---|-----------------------------------------------|-------------------------------------------|
| 1 | Does one thing, or one small family of things | Would match a person across any task      |
| 2 | Can be superhuman at that thing               | Would reason and adapt without retraining |
| 3 | Cannot switch to an unrelated task            | Does not exist today                      |
| 4 | Every deployed system today                   | A research goal, and an open debate       |

**Brilliance at one task doesn't transfer. Ask an unrelated question and the skill is simply gone.**

The test, applied

| What you hear                   | The question to ask                 | Verdict |
|---------------------------------|-------------------------------------|---------|
| "It can write anything"         | Is it one task, or every task?      | Narrow  |
| "It passed a professional exam" | Was it trained for that exam?       | Narrow  |
| "It reasons like a person"      | Across every domain, no retraining? | Narrow  |
| "It's become conscious"         | Is there any evidence at all?       | No      |

## Hype check

A system can be superhuman at one task and helpless at the next. That gap is the whole of narrow AI.

**Which single task is it for? What data was it trained on? That's usually the whole story.**

## 4. The AI already around you

Sub-area 1.4 — everyday artificial intelligence, and where it really shows up.

### The alert before you noticed

Banks have used machine learning on card fraud for years. It is not glamorous, it does not write poetry, and it saves real money every single day.

**Asked:** Watch this card. Learn what this person's normal spending looks like

**Answer:** An unusual payment is flagged in seconds — often before the customer spots it

The same categories, everywhere

| Where        | What it's doing                        | The category       |
|--------------|----------------------------------------|--------------------|
| Your home    | The assistant on your phone hears you  | Speech recognition |
| Your office  | Junk and phishing mail is filtered out | Classification     |
| Your car     | Traffic and fuel range are predicted   | Forecasting        |
| The workshop | A failing part is spotted early        | Anomaly detection  |

## Four jobs that cover most of it

1. Recognition: hearing or seeing what's there
2. Classification: sorting into known groups
3. Forecasting: predicting a number or a trend
4. Anomaly detection: spotting what doesn't belong

## Project Maven

Maven began in 2017 as a narrow programme to help analyse drone imagery. In defence, AI is mostly about attention and speed — not about removing people from decisions.

**Asked:** Analysts were drowning in hours of drone and reconnaissance footage

**Answer:** Machine learning flags what is worth a human's attention — people still make the call

## The habit that matters

Sensors stream readings all day. A model that has learned normal flags the pattern that isn't — and buys the crew time to act.

## The chatbot that cost a company money

The company argued the chatbot was a separate entity. The tribunal disagreed: it is the company's agent. A wrong answer from a machine is still the organisation's wrong answer.

**Asked:** A customer asks the airline's chatbot about bereavement fares

**Answer:** The chatbot gives the wrong answer — and a tribunal in 2024 held the airline responsible for it

## A cautionary tale

In a 2023 New York case, lawyers filed a brief full of invented case citations produced by a chatbot. US courts have been sanctioning lawyers for it ever since, including a Wyoming case in 2025.

**You can't judge a tool you haven't noticed. Noticing is the first act of good judgement.**

**The question is never whether AI is coming. It is whether you can recognise it, and judge it.**

**One family of pattern-finding techniques, nested four deep, all of it narrow, all of it checkable.**

## Domain 1 in six lines

- AI finds patterns in data; it isn't told every rule
- Learning means guessing, comparing, adjusting — millions of times
- AI contains machine learning, which contains deep learning, which contains generative AI

- Narrow AI does one job superbly; general AI doesn't exist yet
- You already use narrow AI daily, from spam filters to fraud alerts
- Fluency isn't understanding — so we always check its work

## Domain 2

# How AI Works (Without the Maths)

Weight 14% · 17 exam questions · about 22 minutes

An accurate, non-mathematical model of how AI produces outputs and where it is unreliable.

**A large language model is a very large predictor of what comes next - it works out what fits, rather than looking anything up.**

## 1. Models, training and inference

Sub-area 2.1, models, training and inference.

**A model holds millions of adjustable settings, tuned on examples until its guesses are good.**

### The training loop

1. Show the model an example
2. Let it make a guess
3. Measure how wrong it was
4. Nudge the settings slightly
5. Repeat, millions of times

### Two phases of a model's life

|   | Training                           | Inference                   |
|---|------------------------------------|-----------------------------|
| 1 | Happens once, before release       | Happens on every request    |
| 2 | Runs for weeks on special hardware | Takes seconds, costs little |
| 3 | The settings move and settle       | The settings are frozen     |
| 4 | It is expensive and slow           | It is what you actually use |

Training and inference

|                  | Training           | Inference          |
|------------------|--------------------|--------------------|
| When             | Once, in advance   | Every request      |
| Dials            | They adjust        | Frozen still       |
| Cost             | Weeks of computing | Seconds, and cheap |
| Learns from you? | From its data      | No, not at all     |

### A model with billions of settings

That is why training is a one-off, industrial-scale job. No conversation could ever move 175 billion dials. The model you meet on Monday is identical on Friday.

**Asked:** Train a language model to predict the next word across an ocean of text

**Answer:** GPT-3, from OpenAI in 2020, ended up with 175 billion settings, tuned on around 570 gigabytes of text

## The practical upshot

Training is far too slow to update daily, so new facts are either placed in the prompt or fetched by a tool.

**Not a tidy list of facts, but a rich statistical sense of which patterns tend to go together.**

## 2. How language models generate text

Sub-area 2.2, how language models generate text.

**The model predicts the next token, appends it, then runs again - so text is built up one piece at a time.**

### The generation loop

1. Read everything so far
2. Score every possible next token
3. Pick one and append it
4. Run again with the new text

**Each step weighs the likelihood of the next piece. The model is completing a pattern, not consulting a database.**

### Watch for this

The model is tuned to produce a plausible continuation, so it can write a confident sentence that simply isn't so.

### The idea that made this work

Attention lets the model weigh how each word relates to the others. It is what turned next-token prediction into fluent, coherent writing.

**Asked:** Keep every word's relationship to every other word in view, however far apart they sit

**Answer:** In 2017, a paper called "Attention Is All You Need" introduced the Transformer - the architecture behind today's language models

**Every reply grew out of one training task: predict the next token. No other objective was ever set.**

**The model also predicts a special stop marker. That is how a reply ends - there, or at a hard limit.**

## 3. Tokens and the context window

Sub-area 2.3, tokens and the context window.

**Text is chopped into tokens - often smaller than a word. They are the units the model counts and works with.**

### How a word gets split

That is OpenAI's own tokeniser, tiktoken. The phrase "tiktoken is great!" comes back as six tokens. As a rule of thumb, one token is about four characters, or roughly three-quarters of an English word.

**Asked:** Give the tokeniser the word "tokenizing"

**Answer:** It comes back as three separate pieces - “token”, “iz” and “ing”

Words, and the tokens behind them

| You think in | The model counts | Rough guide               |
|--------------|------------------|---------------------------|
| Words        | Tokens           | Three-quarters of a word  |
| Characters   | Tokens           | About four characters     |
| A page       | Tokens           | About five hundred tokens |

**The context window is everything the model can see at once - your prompt plus its reply - measured in tokens.**

### The window grew very fast

GPT-4 began at just over eight thousand tokens; its Turbo version reached a hundred and twenty-eight thousand. Even a huge window is a window, not an endless one.

**Asked:** Early language models could hold only a few thousand tokens

**Answer:** By 2024, Claude held 200,000 and Google's Gemini 1.5 Pro held a million - several books at once

### Practical

Once text is trimmed off the desk, the model cannot use it. Restate what is critical and keep key facts close.

**As older turns fall off the desk, replies quietly change - not from mood, but because that text is simply gone.**

## 4. Knowledge cut-off

Sub-area 2.4, the knowledge cut-off.

**Every model is trained on material gathered up to a certain date. After that date, it has nothing to draw on.**

**Training takes weeks or months. By the time a model is finished, the world has already moved on.**

### Real cut-off dates

These dates are published, model by model, and they move with every new version. So the question about any tool you rely on is not “is it clever?” but “where does its knowledge stop?”

**Asked:** Ask a model what it knows, and the edge shows up as a date

**Answer:** The original GPT-4 was trained to September 2021; its Turbo version reached December 2023

What the cut-off does and does not mean

| Question               | Answer                |
|------------------------|-----------------------|
| Is it a memory?        | Patterns, not files   |
| Is older data safe?    | Still can be wrong    |
| Can it know late news? | Only if supplied      |
| How do you fix it?     | Give it current facts |

## Common trap

The cut-off is the end of the training data, not today's date. Anything more recent must be supplied, or the model may fill the gap.

## Two ways it hides the edge

You cannot tell the two apart by the confidence, because both sound certain. The second failure is the dangerous one, since nothing in the reply signals that the ground has run out.

**Asked:** Ask about something that happened after the cut-off

**Answer:** It either says it cannot help, or it answers anyway - fluently, and wrongly

**Supply the facts yourself, or use a tool that fetches them. Never treat an unverified recent claim as current.**

## 5. Variability and embeddings

Sub-area 2.5, variability and embeddings.

**At each step the model samples from a spread of likely next tokens, so wording and detail can shift every run.**

### The randomness knob

It is usually called temperature. Lower settings give steadier, more repetitive answers. Higher settings give more variety, and a little more risk.

**Asked:** Most tools expose a setting for how adventurous the model is

**Answer:** Turned right down, it nearly always takes the front-runner; turned up, it roams further

### Worth remembering

It is why these tools read as human. But it also means no single answer is a fixed record - treat output as a draft.

**An embedding turns meaning into a list of numbers, arranged so that similar meanings sit close together.**

### Meaning you can do arithmetic with

Google researchers showed this in 2013 with a technique called word2vec. The same arithmetic - "Paris" minus "France" plus "Italy" - lands on "Rome".

**Asked:** Take the numbers for "king", subtract "man", add "woman"

**Answer:** The nearest word to the result is "queen"

### What embeddings make possible

1. Turn the query into a position
2. Turn each document into a position
3. Find the nearest ones
4. That is meaning-based search

**Similar vectors mean similar usage, not identical meaning. Opposites can end up surprisingly close together.**

**A frozen predictor of the next token, working inside a finite window, up to a fixed date, producing a fresh answer each time.**

### **Domain 2 in six lines**

- A model is millions of settings, tuned by training, then frozen
- Inference is simply using it, and it never learns from your chat
- Text is generated one token at a time, by predicting what comes next
- Tokens fill a finite context window - what falls off the desk is gone
- Every model has a knowledge cut-off; newer facts must be supplied
- Sampling causes variability; embeddings turn meaning into numbers

## Domain 3

# The AI Landscape & Generative AI

Weight 12% · 14 exam questions · about 20 minutes

A vendor-neutral map of AI capabilities and how to choose the right category of tool.

**The landscape is a map, not a maze. Every tool can be placed by four questions.**

## 1. Generative AI and output types

Sub-area 3.1, Generative AI & Output Types.

**Generative systems create new content rather than sorting what you give them. The output type decides where a tool earns its place.**

### Four kinds of output

1. Text: prose, summaries, translations
2. Images: pictures, diagrams, mock-ups
3. Audio: speech, music and sound effects
4. Code: working software in a chosen language

### Text, produced one word at a time

It is not looking anything up. The words are generated, one at a time, from patterns absorbed during training. That is why it can sound right and be wrong in the same sentence.

**Asked:** Write a short, polite email asking for more time on a deadline

**Answer:** A fluent draft appears in seconds - not because it knows you, but because it has seen millions of similar emails

### Pictures, built out of noise

Most image generators today are diffusion models. In 2022 the category opened up, including one released openly, with its weights published for anyone to run - so it was no longer locked inside a few laboratories.

**Asked:** Describe a scene, a diagram or a style, in ordinary words

**Answer:** A usable first-draft image in seconds - made by removing noise, step by step, until a picture matches the description

### Sound that can fool the ear

Speech synthesis crossed a threshold in 2016, when a system called WaveNet produced strikingly human voices. In 2019, a British energy company was reported to have lost around two hundred thousand pounds to a call from a cloned voice of its own chief executive.

**Asked:** A short recording of someone speaking

**Answer:** Any voice, cloned and made to say almost anything - which is powerful for good, and dangerous for obvious reasons

### Software that writes software

Code assistants arrived around 2021, built on language models. Code that runs is not the same as code that is correct or safe: it will do what you asked, which may not be what you meant.

**Asked:** Tidy this spreadsheet, fix this error, write this small function

**Answer:** Working code appears in the language you name - from the same language-model foundations as the writing tools

## The common trap

Generative content arrives looking complete. That polish belongs to the format, not to the content, and it is no proof of correctness.

## 2. Foundation models and multimodal AI

Sub-area 3.2, Foundation Models & Multimodal AI.

**Trained once on a very large body of data, then reused across many tasks. It is the groundwork that many products stand on.**

**The best-known foundation models work mainly with text. Scale is the point: at size, ability appears that was never specifically taught.**

**You rarely use the base directly. You adapt it - sometimes by fine-tuning, sometimes just by how you ask.**

**A modality is a kind of input or output. A multimodal system accepts more than one kind of input, and can answer in more than one kind of output.**

### How the join was made

Around 2021, a system called CLIP learned to link images and their descriptions. Speech followed, when models like Whisper could transcribe audio directly by 2022. First text, then pictures, then sound.

**Asked:** Hundreds of millions of pictures, each with its own caption, from the open internet

**Answer:** A system that has learned that a photo of a dog and the word dog belong together - so you can ask it about a picture in plain language

Inputs and outputs

| Kind       | Can accept             | Can produce            |
|------------|------------------------|------------------------|
| Text-only  | Typed words            | Written words          |
| Vision     | Words and pictures     | Written words          |
| Multimodal | Words, pictures, sound | Words, pictures, sound |

## Precision

When two tools behave almost identically, they may share a foundation model underneath. So you are really choosing the wrapper - the interface, the safeguards, the data handling.

## 3. Retrieval-augmented generation

Sub-area 3.3, Retrieval-Augmented Generation (RAG).

**A plain model answers from training memory, which can be stale. When it does not know, it does not usually say so - it fills the gap with something plausible.**

### How RAG works

1. Your question is used to search a trusted source

2. The best-matching passages are retrieved
3. Those passages are added to the request as context
4. The model writes its answer from them

**A plain model answers from training memory. RAG answers from a source you can inspect, sitting right there in front of you.**

### Same question, two routes

One answer quietly invites trust. The other invites verification. The retrieval tool has not become cleverer - it simply has the document in front of it.

**Asked:** Summarise the current policy on this subject

**Answer:** A plain model invents a plausible clause; a retrieval tool quotes the actual clause from the actual document, which you can then open and check

**Retrieval usually relies on embeddings, which place similar passages close together. But it can only retrieve what is in the source, and it can retrieve the wrong passage.**

### The discipline

A retrieval tool points you at a source, which is a great gift - but you still have to look. Ask what it retrieved, and check the passage is current.

**It is how an assistant can answer from your organisation's own documents, quoting a policy rather than guessing at it. Keep a trusted source, search it, and put the matches in front of the model.**

## 4. AI agents

Sub-area 3.4, AI Agents.

**Given a goal, it plans steps, uses tools to carry them out, checks the result and adjusts - then keeps going until the goal is met. The key word is independence.**

Chatbot or agent?

|             | Chatbot          | Agent             |
|-------------|------------------|-------------------|
| You give it | A question       | A goal            |
| It returns  | An answer        | Completed steps   |
| It acts     | No               | Yes, using tools  |
| Your role   | Check the answer | Check the actions |

**The system thinks about the next step, does it, observes what happened, and thinks again - a loop of think, act, observe, repeat.**

### The year agents went public

It was exciting, and at first unreliable: it wandered, repeated itself and ran up costs. The honest picture: capable at narrow, well-scoped jobs with the right tools; not yet dependable at open-ended ones.

**Asked:** Give it a goal in plain language - no step-by-step instructions

**Answer:** It breaks the goal into tasks, runs them using tools, and keeps going on its own - the idea behind the 2023 open-source project AutoGPT

**An agent that can act can also act wrongly, and many times before anyone notices. Keep it on a short leash: narrow permissions, human approval for anything irreversible.**

## A cautionary tale

In 2023, a US car dealership's chatbot was asked, jokingly, whether a car could be bought for a dollar. It agreed, and offered to seal the deal. A goal without guardrails will chase it to the edge.

**An agent earns its place when the path is genuinely uncertain and the steps depend on what is found along the way. Reaching for one when a simpler tool would do adds risk, not value.**

## 5. Model, application and service

Sub-area 3.5, Model, Application & Service.

### Three layers

1. A model: the trained system that predicts
2. An application: the product you actually use
3. A service: a model offered over a network
4. Each layer stands on the one below it

Who controls what

| Layer       | What it is              | Who controls it |
|-------------|-------------------------|-----------------|
| Model       | The trained system      | Its developer   |
| Service     | Access over a network   | Its provider    |
| Application | The product you use     | Its vendor      |
| Your use    | Your task and your data | You             |

**The model is the engine. The application is the finished car. The service is the hire network that lends that kind of car to anyone who needs one.**

**When you type into an application, your words may travel across a network to a service run by somebody else - which is exactly where data-handling questions live.**

**Some models run entirely on your own device; others live in a data centre, or on your organisation's own machines. That choice is a security choice.**

### Precision

Two applications sharing a foundation model may behave alike - but their terms, safeguards and data handling can be worlds apart. Judge the layer you can actually see.

### Following the chain

The application is the screen, the service is the network, the model is the engine - and each layer may belong to somebody different.

**Asked:** You ask an assistant on your phone about a document you have uploaded

**Answer:** An application takes your words, a service does the work over the network, and a model produces the answer - three layers in one ordinary interaction

## 6. Selecting a tool category

Sub-area 3.6, Selecting a Tool Category.

**Describe the job, name the output you need, decide what it must read and how much it may do alone - then reach for a tool. Task first, category next, product last.**

### Four questions that name the category

1. What must it produce? The output
2. What must it read? The input
3. Must it use a specific source? Grounding
4. Must it act on its own? Autonomy

Task to category

| Your task          | You need      | Category        |
|--------------------|---------------|-----------------|
| Draft some prose   | Text output   | Text generator  |
| Work from a policy | A real source | Retrieval tool  |
| Describe a photo   | Pictures in   | Multimodal tool |
| Repeat a small job | Action        | Agent           |

**Paying for capability you will never use, and asking a tool to do something it was never built for. Both are avoided by matching category to task.**

**Two things narrow the choice beyond the output: how sensitive the information is, and how hard the answer is to verify.**

### Good practice

What must it produce? What must it read? Must it use a specific source? Must it act on its own? Those answers name the category - and only then is it worth comparing products.

**Get the category right and almost any competent product in it will do the job. Get the category wrong and no product, however good, will save you.**

**Generative tools differ by output, by input, by how you reach them, and by how much they do alone. Name the category and you have done most of the thinking.**

### Domain 3 in seven lines

- Generative AI creates text, images, audio or code
- Foundation models are trained once and reused widely
- Multimodal means many inputs and many outputs
- RAG answers from a source you can inspect
- An agent acts on a goal, not just an answer
- Model, application and service are three layers
- Choose the category before the product

## Domain 4

# Prompt Engineering Fundamentals

Weight 16% · 19 exam questions · about 24 minutes

The highest-leverage practical skill: constructing effective, reliable prompts.

**A prompt is a specification, not a question. Say what you want, for whom, in what shape, and within what limits.**

## 1. Anatomy of a prompt

Sub-area 4.1, anatomy of a prompt.

**Task, context, format and constraints — the same four parts, every single time.**

### A weak prompt, in full

No reader, no purpose, no shape, no limits. So the model fills the silence with vaguely true padding that could have been written for anyone.

**Asked:** Write something about fuel handling.

**Answer:** Fuel handling is an important subject. There are many kinds of fuel, including petrol, diesel and aviation fuel. When handling fuel, it is important to follow the correct procedures and stay safe at all times. Requirements vary, so you should always check your local guidelines.

### The same task, prompted well

Same subject. Same model. The only thing that changed was the specification — a named role, a named reader, a shape, and a limit.

**Asked:** As a training officer, write a one-page brief on safe fuel handling for new drivers with no technical background. Use five short sections with headings, plain English, and finish with three safety checks they must remember.

**Answer:** Five short sections with headings, in plain words — and a list of three safety checks at the end, exactly where a new driver would look for them.

### What actually changed

|   | The weak prompt       | The strong prompt        |
|---|-----------------------|--------------------------|
| 1 | One vague instruction | A clear action named     |
| 2 | No reader named       | Reader and purpose set   |
| 3 | No shape given        | Shape and headings fixed |
| 4 | No limits set         | Length and tone limited  |

**The famous “trick” prompts work for the same reason every prompt works: a clear task, context, format and limits, in the right order.**

### Building one from scratch

1. Name the task in one clear verb
2. Add the context and the reader
3. Say what shape the answer takes

4. Set the limits: length and tone

## Rule of thumb

Write the prompt as if briefing a capable new starter who knows nothing about your particular task.

## 2. Role and audience

Sub-area 4.2, role and audience.

**A role tells the model where to answer from — the viewpoint of a trainer, an engineer, a policy adviser.**

**Name the reader and the depth fixes itself — same question, two readers, two quite different answers.**

### Role and audience do different jobs

|   | Role                  | Audience                  |
|---|-----------------------|---------------------------|
| 1 | Sets the perspective  | Sets the depth            |
| 2 | Answers “from where?” | Answers “for whom?”       |
| 3 | Example: as a trainer | Example: for new recruits |
| 4 | Changes the voice     | Changes the detail        |

### With a role, and without

Nothing about the facts changed. What changed was where the answer came from, and who it was for.

**Asked:** Explain the new reporting deadline. ...then... As a team leader, explain the new reporting deadline to a new joiner who's nervous about getting it wrong.

**Answer:** The first reply is flat and neutral — correct, but cold. The second is warmer and plainer, and it leads with reassurance before the detail.

**It knows nothing of your team, your deadline or your reader unless your prompt says so.**

### In practice

“As a training officer, write a one-page brief on fuel handling for new drivers who have no technical background.” One sentence, four choices made.

## 3. Few-shot prompting

Sub-area 4.3, few-shot prompting.

**An example is a specification you don't have to describe. The model reads the pattern straight off it.**

### Zero, one, few

1. Zero-shot: no examples, just the instruction
2. One-shot: a single worked example
3. Few-shot: two or three examples
4. The pattern, tone and format come from the examples

## Teaching a format by example

The single example has already shown the shape, the order and the level of detail. The model reads the pattern and fills in the rest.

**Asked:** Item: ration pack | Condition: serviceable | Action: reorder — continue this table for the items I've listed.

**Answer:** Item: water can | Condition: damaged | Action: replace · Item: boots | Condition: serviceable | Action: none

Match the examples to the job

| Approach  | Examples given | Best for                  |
|-----------|----------------|---------------------------|
| Zero-shot | None           | Simple, familiar tasks    |
| One-shot  | One example    | Fixing tone or style      |
| Few-shot  | Two or three   | Precise format or pattern |

**The things you'd never think to write down — the small conventions — are exactly what an example passes on.**

**A good example and a bad one, side by side, teach the boundary faster than any description.**

### Worth knowing

If your examples disagree with one another, the model faithfully copies the confusion. Keep them tight, realistic and alike.

## 4. Structure and decomposition

Sub-area 4.4, structure and decomposition.

**Decomposition trades one enormous prompt for a short chain of small, checkable ones.**

### A chain of small, checkable steps

1. Ask for the outline
2. Agree the shape
3. Draft one section at a time
4. Check each before the next
5. Assemble the finished whole

**Get the outline approved before a single section is written. It's the cheapest correction you'll ever make.**

Match the approach to the task

| Task size          | Approach               |
|--------------------|------------------------|
| One simple action  | A single clear prompt  |
| A few linked steps | A short sequence       |
| A large document   | Outline, then sections |
| Anything long      | Check each step first  |

**A good step is reusable. Keep the ones that worked and run them again on the next job.**

**A chain of small steps is easier to reuse, and much easier to trust.**

### Good practice

If a prompt contains the word “and” three times, it's probably three prompts wearing one coat.

## 5. Iterative refinement

Sub-area 4.5, iterative refinement.

**You can't specify everything at once. Refinement is how you discover what you actually wanted.**

### The refinement loop

1. Read the first answer as a draft
2. Name one thing to change
3. Say what to keep, as well
4. Repeat until it's fit for purpose

### Refining by the turn

Not “make it better”. Point at the change you want, precisely, and let everything else stay where it is.

**Asked:** Make the second paragraph shorter and drop the jargon.

**Answer:** The same announcement returns with exactly that paragraph tightened and the technical terms replaced — everything else untouched, because you said what to keep.

**Refinement works because the model can still see the draft you're talking about.**

### Worth knowing

Change “make it better” into “cut it to half the length and use plain English”. Vague feedback earns a vague rewrite.

**Refinement ends when the draft is fit for purpose, not perfect. Past a point, more turns stop helping.**

## 6. Common prompting faults

Sub-area 4.6, common prompting faults.

**Vague task, no format, missing context, unverified output, and too many tasks at once.**

#### Fault and fix

| Fault             | Fix                     |
|-------------------|-------------------------|
| Vague instruction | Name the exact task     |
| No format         | State the shape         |
| Missing context   | Supply the background   |
| Unverified output | Check before you use it |
| Overloaded prompt | Split into steps        |

**Vague task, missing context, no format — they all come down to leaving the model to guess.**

#### All the faults in one line

Vague task, no format, no reader, two unrelated jobs jammed together with “and”, and a tone instruction bolted on as an afterthought.

**Asked:** Tell me some ideas about our training and also summarise the policy and make it formal.

**Answer:** A muddled answer that is half scattered ideas and half a policy summary, formal in a strange, padded way — and neither job done well.

#### The same task, fault-free

Role, task, audience and limit, all named. No “and”-stacking, no second job, no vague “some ideas”.

**Asked:** You are a training lead. Write three short bullet ideas for improving our weekly safety briefings, for the briefing team, each under twenty words.

**Answer:** Three tight, usable suggestions — each one short, concrete, and ready to act on immediately.

#### Common trap

The worst output is fluent, plausible and false. Never let good grammar convince you that a fact is correct.

**No prompt, however clever, removes your duty to check. Verify at source, then use.**

**A good prompt specifies the task, the context, the format and the constraints — then you refine it, turn by turn.**

#### Domain 4 in six lines

- A prompt is a specification: task, context, format and constraints
- Role sets the perspective; audience sets the depth
- Few-shot: show two or three examples of the pattern
- Decompose big tasks into small, checkable steps
- Refine over turns — the first answer is only a draft
- Never trust unverified output — always verify at source

## Domain 5

# Working with AI Day to Day

Weight 12% · 15 exam questions · about 20 minutes

Practical, safe daily workflows — drafting, summarising, ideation and the verification habit.

**The tool drafts. You decide, and you own it.**

## 1. Drafting and rewriting

Sub-area 5.1, drafting and rewriting.

**Hand over your own notes and points, then ask for a first draft. You supply the substance; it tidies the shape.**

### The drafting loop

1. Brief it with your own notes and points
2. Ask for a first draft, not a final one
3. Cut, correct, and add what only you hold
4. Read it aloud, then put your name to it

### The routine memo

The model improves the form. It cannot supply the figure from last week's return, or the reason this particular decision was taken. Those live in your head, or in the file.

**Asked:** A rough set of notes, plus the audience: a short routine memo for the chain of command

**Answer:** A tighter, better-ordered draft in seconds - but two facts are still missing

### Who supplies what

|   | You supply                 | It supplies         |
|---|----------------------------|---------------------|
| 1 | The facts and the figures  | A clean structure   |
| 2 | The judgement and the tone | Tighter sentences   |
| 3 | The accountability         | A much faster start |

### Common trap

A tidy draft can read perfectly and still say nothing distinctive. Guard the detail only you can supply.

### Turning jargon into plain English

Rewriting is where the tool earns its keep. But read the result aloud - if the words no longer sound like you, treat that as a warning, not a win.

**Asked:** "It is recommended that stakeholders be engaged to facilitate the optimisation of the deliverable prior to finalisation."

**Answer:** "Check with the people affected before we finish it."

## 2. Summarising

Sub-area 5.2, summarising.

**Compress it well and you save time. Compress it badly and your reader inherits your error - under your name.**

### How to summarise safely

1. Give it the whole document, not one paragraph
2. Ask for the shape you need: lines, or a short list
3. Ask a separate question about caveats and conditions
4. Read the result against the original

### The caveat that vanished

A summary can keep the recommendation and lose the condition attached to it. That single loss can change the meaning completely.

**Asked:** A paragraph on a new policy, with one key sentence in the middle: “this applies only to permanent staff”

**Answer:** A clean summary that reads “the policy applies to staff” - the condition has quietly disappeared

What a summary tends to drop

| In the document  | Often lost         | Keep it      |
|------------------|--------------------|--------------|
| A recommendation | Its conditions     | Restore them |
| A figure         | Its date or source | Cite both    |
| A caveat         | The whole caveat   | Quote it     |
| A duty           | Who is responsible | Name them    |

### Careful

In a policy or a set of orders, the condition is the content. Never let a summary travel on its own.

### From long report to short briefing

The model gives you the shape. You re-read the source and put back what it dropped. That is the standard to work to.

**Asked:** A forty-page report, and one page of space for the morning briefing

**Answer:** A clean summary in seconds - then the analyst restores the two caveats it left out

## 3. Ideation, planning and research

Sub-area 5.3, ideation, planning and research assistance.

**Ask for many options, then choose. The tool has never met your people, your constraints or your problem.**

### Widen, then choose

1. Ask for many options, not just one
2. Ask it to argue the opposite case
3. Group and prune to the few that matter
4. Test the survivors against reality yourself

### Ten ideas, three kept

The tool is generous with ideas, which is exactly its strength. The pruning, the judgement and the decision stay with you.

**Asked:** A problem with no obvious solution, and a request for ten possible approaches

**Answer:** Ten suggestions - you group them, and keep the three worth costing out

## Context

Treat its facts as leads, not answers. Ask for directions, then go and read the source yourself.

## A lead, not an answer

The tool is often good at pointing towards a subject, and unreliable on the detail. Go to the source, and check what it actually says.

**Asked:** "Find me a study on how long this process usually takes."

**Answer:** It names a plausible-sounding report and a neat statistic - which you then cannot find when you open the real study

## Planning a training day

The model gives the shape of the day. The team supplies the facts - the room, the budget, the availability. Neither alone finishes the job.

**Asked:** "Suggest a running order for a one-day training event."

**Answer:** A sensible shape with the breaks in the right places - then the team checks the venue, the timing and the cost

## 4. Appropriate use

Sub-area 5.4, appropriate use.

### What you ask it to do, and what you put into it. Both halves have to be right.

## The Samsung leak

In 2023, Samsung found that data sent to a public chatbot sits on someone else's servers, is hard to retrieve, and may be reused. The tool was convenient; the cost was confidential code out of their control.

**Asked:** Engineers asked a public chatbot to help fix faulty code, pasting in confidential source code and internal meeting notes

**Answer:** The data left the building - so Samsung banned generative tools on company devices and built its own internal system

## Whose servers, whose rules

Italy's privacy regulator, the Garante, restricted ChatGPT over how it handled personal data. A public service has its own retention and training rules - and they are not yours to assume.

**Asked:** A public chatbot gathers a vast amount of users' personal data in order to train its models

**Answer:** In 2023, Italy's data authority briefly blocked it, finding no proper legal basis for that collection

Where the line usually sits

| The task           | Sensible use    | Keep it human  |
|--------------------|-----------------|----------------|
| Draft a message    | A first pass    | Tone and facts |
| Summarise a policy | A rough shape   | The caveats    |
| Weigh a decision   | Explore options | The decision   |
| Personal data      | Not at all      | Entirely       |

## Common trap

A tool being easy to reach doesn't make the task allowed. Check the rules before you paste, not after.

**No general tool's suggestion outranks your own policy. Where the two conflict, the policy wins.**

## 5. The verification habit

Sub-area 5.5, the verification habit.

**The tool outputs a draft. The moment it leaves your hands, it becomes your claim.**

### The cases that never existed

In the case of *Mata v. Avianca*, the model even insisted the fake cases were real and findable on legal databases. Courts have kept sanctioning lawyers for exactly this ever since.

**Asked:** A lawyer asked a chatbot to find cases supporting his argument, and filed the citations it gave him

**Answer:** Six of them were invented - with fake names, dates and judges. In 2023 the court fined the lawyers and their firm

What to check, and against what

| Check this           | Against this         |
|----------------------|----------------------|
| Facts and figures    | The primary source   |
| Any arithmetic       | Your own calculation |
| Citations            | The document itself  |
| Names, dates, titles | The record           |

## The four checks

1. Every fact: trace it to a source you trust
2. Every sum: do it once more yourself
3. Every citation: open it and read it
4. Every name and date: check the record

## Good practice

A low-risk note needs a quick read. Anything that goes on record deserves a second pair of eyes.

### The sum that looked fine

Language models predict plausible text, and a plausible number can still be the wrong number. Re-doing the arithmetic takes a moment and removes the risk.

**Asked:** “Add up these figures and give me the total.” A request the tool answers instantly and fluently

**Answer:** The total is wrong by a small amount - plausible enough to slip through if nobody re-adds the column

## The date that drifted

Dates and names are where quiet errors love to hide, and they are also the easiest things to check. A confident answer is not a checked one.

**Asked:** “When did this regulation come into force?” One fact, answered instantly and without hesitation

**Answer:** A confident date that turns out to be the day the consultation opened, not the day the rule took effect

## 6. The personal prompt library

Sub-area 5.6, the personal prompt library.

**When a prompt gives a reliable result, save the exact wording. Good work shouldn't depend on your memory.**

### A four-line entry

1. The prompt wording, kept exactly as it worked
2. A note on what you changed, and why
3. The task it serves, so you can find it
4. A review date, so it stays current

### A prompt worth keeping

The saved wording is what makes the result repeatable. The note is what makes it improve.

**Asked:** “Rewrite this in plain English, keep it under eighty words, and keep the technical terms that must stay exact.”

**Answer:** Saved verbatim and labelled “plain-English rewrite”, with a note that the length limit stopped the output rambling

What a good entry holds

| Keep               | Why                 |
|--------------------|---------------------|
| The prompt wording | So results repeat   |
| A note on changes  | So you improve it   |
| The task it serves | So you can find it  |
| A review date      | So it stays current |

## Good practice

A prompt that suits your work may not suit the person beside you. Pass on the thinking, and let them adapt the words.

**You don't need a fresh start. The prompts you already use are the first entries in your library.**

## Two ways to keep your best work

|  |                 |                 |
|--|-----------------|-----------------|
|  | Scattered notes | A small library |
|--|-----------------|-----------------|

|   |                           |                             |
|---|---------------------------|-----------------------------|
| 1 | Work redone from scratch  | The same result, repeatable |
| 2 | Good prompts forgotten    | Each prompt a little better |
| 3 | Results that vary by luck | A method you can explain    |
| 4 | Nothing to build on       | Your own working knowledge  |

**Aid your thinking, keep your judgement, and check before it leaves your hands.**

### **Domain 5 in six lines**

- Draft with the tool; supply the substance and keep your voice
- A summary drops caveats - restore them from the original
- Widen your ideas; never let it decide, and never let it be your source
- Appropriate use: keep private material out, and check the rules first
- Verify facts, figures, sums and citations before they leave
- Keep a small library of prompts you trust

## Domain 6

# Limitations, Risks & Hallucination

Weight 16% · 19 exam questions · about 22 minutes

Why fluent output can still be wrong, and how to reduce the risk.

**Fluent does not mean factual. A language model predicts plausible words; it does not verify them.**

## 1. Hallucination

Sub-area 6.1 — hallucination.

**Output that is fluent, confident and wrong — with no hesitation and no warning sign.**

**The model predicts the next likely word. Likely is not the same as true.**

### A confident answer, entirely invented

Nothing in the tone warns you. The answer matches the shape of your question, not the facts of the world.

**Asked:** Ask about a regulation you only half-remember, and give it no number

**Answer:** Back comes a regulation number, a date and a tidy summary of its provisions lu2014 all invented to fit the question

**Lying needs intent. A hallucination has none — it is a pattern being completed, not a deception being told.**

### The trap

There is no flicker, no hedge, no change of tone when the model runs out of truth and starts inventing.

### Two things that feel identical

|   | Sounds right                        | Is right                      |
|---|-------------------------------------|-------------------------------|
| 1 | Correct grammar and fluent phrasing | Checked against a real source |
| 2 | A confident, assured tone           | Verified by a person          |
| 3 | Neat, tidy and well-formatted       | Traceable to real evidence    |
| 4 | Not a hint of hesitation            | Something you can defend      |

**Statistics, dates, quotations, names and references — the very things people cite to sound credible.**

## 2. Stale information and fabrication

Sub-area 6.2 — stale information and fabrication.

**Training stops at a fixed date. Anything after it, the model has never seen.**

**Asked about the present, answered from the past**

It is not lying to you. It has no way of knowing what happened after its cut-off, so it answers with the last version it ever saw.

**Asked:** Ask for this year's allowance, or this month's price, without handing over a current source

**Answer:** It answers confidently from its training data \u2014 even if the figure changed months or years ago

**The faster a field moves, the more of a model's knowledge has gone out of date by the time you ask.**

**Stale information is out of date. Fabrication was never true at any time.**

### The commonest fabrication

Author, title, publication, year, page numbers. A fabricated citation has every field filled in \u2014 and none of it exists.

### Mata v. Avianca — the case that wasn't there

The most quoted cautionary tale in the field, and completely real: fluent, confident, well-formatted citations to cases that never existed.

**Asked:** A lawyer asks a chatbot for supporting cases, then files them in a New York court brief

**Answer:** The cases were invented \u2014 with fake names, judges and dates. The lawyers were sanctioned in 2023, and US courts kept sanctioning lawyers for AI-fabricated citations through 2025

### Stale against fabricated

|   | Stale information               | Fabrication                   |
|---|---------------------------------|-------------------------------|
| 1 | Was true at some point          | Was never true at all         |
| 2 | Out of date now                 | Invented on the spot          |
| 3 | Correctable with a fresh source | Only fixable by discarding it |
| 4 | Example: last year's figure     | Example: an invented citation |

## 3. Provenance and citation checking

Sub-area 6.3 — provenance and citation checking.

**The trail behind a statement. Where did this come from, and can you follow that trail yourself?**

### Four questions for any claim

1. Where did this come from?
2. Can I reach that source myself?
3. Does the source actually say what is claimed?
4. Is the source current and authoritative?

### Checking one citation, start to finish

You did not need legal training to catch it. You needed one search for the title, and one attempt to open the source. That is the whole check.

**Asked:** The model cites: Smith v. Jones, 2019, volume 42 of a law report, page 118 \u2014 a tidy, complete-looking citation

**Answer:** Search the exact title and the two parties. No such case. Try to open volume 42, page 118. No such page. It was assembled to look right, and nothing behind it exists

Checking a citation, step by step

| Step      | What you do                | What it catches        |
|-----------|----------------------------|------------------------|
| Locate    | Search the exact title     | An invented source     |
| Open      | Read the original          | A misquoted passage    |
| Date      | Check the publication date | Stale information      |
| Attribute | Confirm the author         | A fake or wrong author |

**A claim is only as good as the source you can reach and read for yourself.**

**A search for the title costs seconds. Finding out in front of an audience costs far more.**

### The rule of thumb

If you cannot get to the original yourself, you have a sentence that reads well \u2014 not a verified fact. They are not the same thing.

## 4. Reducing the risk

Sub-area 6.4 — reducing the risk.

**Hallucination is a property of the method, not a setting. The lever you control is how you use it.**

### Five measures that genuinely help

1. Ground it: supply the source material yourself
2. Quote it: ask for the exact sentence, then check it
3. Question it: ask what it is unsure about
4. Date it: use a live search for anything current
5. Own it: keep a person on the final decision

The measures, and what each one cuts

| Measure   | In practice                 | Reduces       |
|-----------|-----------------------------|---------------|
| Ground it | Supply the sources yourself | Invention     |
| Quote it  | Request the exact wording   | Misquoting    |
| Date it   | Check the publication date  | Staleness     |
| Verify it | Open the source yourself    | False sources |
| Own it    | A named person decides      | Unchecked use |

### Grounding it in your own document

Grounding does not make the model truthful. It makes it accountable to a page you already hold, which you can open and check yourself.

**Asked:** Give the model the contract, the policy or the manual, and ask it to answer only from that document

**Answer:** Its job changes from remembering to reading \u2014 and you can check every answer against the page it came from

**None of these measures require understanding the maths. They require a habit of checking.**

## Good practice

A draft is something you improve and check. Evidence is something you verify. Do not confuse the two.

**Every measure points the same way: a person checks the claim before it is relied on.**

## 5. Where verification is mandatory

Sub-area 6.5 — mandatory-verification contexts.

**Low-stakes drafting gives you room to judge. High-stakes output gives you no such room.**

## Verify before you use

1. Anything feeding a decision or a formal record
2. Safety, legal, medical and financial matters
3. Anything published, briefed or passed onward
4. Statistics, figures, dates and quotations
5. Anything attributed to a person or an organisation

## Where the accountability lands

In high-stakes settings, 'a machine produced it' is not a defence. The output is the organisation's output, and so is the error.

**Asked:** A chatbot speaks for a company and gets a customer's entitlement wrong

**Answer:** A Canadian tribunal in 2024 held the company responsible \u2014 the chatbot is the company's agent, and the company owns what it says

**A human source check, and a named owner, before the output moves anywhere.**

## If in doubt

In any formal setting, an unattributed machine claim carries no weight at all. Verify it, or leave it out.

Where the check is mandatory

| Context                 | Why the stakes rise          | The check                   |
|-------------------------|------------------------------|-----------------------------|
| A decision or record    | The error becomes the record | A human source check        |
| Safety, legal, money    | Harm is real and lasting     | A qualified person verifies |
| Anything briefed onward | The error travels with it    | Verify before you pass it   |
| Figures and quotations  | Invented most fluently       | Open the original source    |

**Before a claim moves: could I stand behind this if I were asked, right now, where it came from?**

**Fluent, confident output can still be wrong. Check the source, or don't rely on it.**

## **Domain 6 in six lines**

- Hallucination is fluent, confident and false
- Stale information is out of date, not invented
- Fabrication, including invented citations, was never true
- Check provenance: locate, open, date, attribute
- Ground it, quote it, date it, verify it, own it
- High-stakes work always needs a human source check

## Domain 7

# AI Security, Privacy & Data

Weight 10% · 12 exam questions · about 18 minutes

Protecting information when AI services are in the loop — data handling, shadow AI, and AI-enabled threats.

**An AI service treats whatever you give it as data it may keep, learn from, or repeat.**

## 1. Protecting information

Sub-area 7.1, protecting information.

**Some information must never be entered into a public AI service. Not sometimes. Never.**

### Two honest questions

|   | Safe to enter                    | Never enter                   |
|---|----------------------------------|-------------------------------|
| 1 | Information already published    | Anything carrying a marking   |
| 2 | Your own rough notes             | Personal data about anyone    |
| 3 | Nothing that names a real person | Credentials, plans or timings |

### Common trap

Three unclassified details can combine into something sensitive. The service keeps all the pieces.

**A protective marking exists because somebody judged that this would harm the wrong hands. Respect the judgement.**

The protective ladder

| Level              | What it means       | Public tool?       |
|--------------------|---------------------|--------------------|
| Unmarked           | Never been assessed | Ask, do not assume |
| Official           | Routine business    | Often still no     |
| Official sensitive | Would cause harm    | Never              |
| Higher still       | Serious damage      | Never, ever        |

### Four questions before you paste

1. Is it already public?
2. Does it name or identify anyone?
3. Would harm follow from a leak?
4. Does it carry a marking?

**Nobody is expected to memorise the rules. You are expected to ask when you are unsure.**

## 2. Data handling by AI services

Sub-area 7.2, data handling by AI services.

**Your input can be used to train the model, stored for years, read by reviewers, or shared with suppliers.**

### The journey of one sentence

1. You type it, and it is yours
2. You press send, and it is captured
3. The service stores or logs it
4. Depending on the tier, it may train the model
5. Third parties may process or review it

Who can touch it

| Stage          | What happens      | Under your control?  |
|----------------|-------------------|----------------------|
| You type it in | Input captured    | Yes, until you send  |
| The service    | Stored or logged  | Partly, via settings |
| Training       | Feeds the model   | Only if offered      |
| Third parties  | Process or review | Rarely at all        |

### Know your tier

Consumer tools, business tiers and self-hosted systems have very different defaults. Check which one you are on, and read what its terms actually say.

**Input used for training can resurface in answers given to strangers. Your sentence does not stay yours.**

**Changing a name does not make personal data safe. The service still holds everything around it.**

### The chatbot the company owned

Moffatt v. Air Canada is the case. The company argued the chatbot was a separate thing. The tribunal disagreed. A chatbot is the organisation's agent, so the organisation owns its words.

**Asked:** A passenger asks a company's chatbot about a bereavement fare

**Answer:** It answers wrongly, and in 2024 a Canadian tribunal held the company responsible for what it said

## 3. Shadow AI

Sub-area 7.3, shadow AI.

**Shadow AI is any use of an artificial intelligence tool your organisation has not approved. It is everywhere.**

### What it actually looks like

1. A browser extension that summarises pages
2. A personal account used to draft a work email
3. A free tool chosen because the approved one was down
4. A document pasted into a chat to beat a deadline

### Approved route or shadow route

|   | Approved tool                  | Shadow tool                    |
|---|--------------------------------|--------------------------------|
| 1 | Somebody vetted it             | Nobody vetted it               |
| 2 | Usually logged                 | Not logged                     |
| 3 | Data handling is known         | Data handling unknown          |
| 4 | If it leaks, you can report it | If it leaks, nobody would know |

**When approved tools go to some and not others, the rest improvise. That improvisation is shadow AI.**

**Clear guidance, a sensible default, and a route that is genuinely easier than the shadow one.**

### Good practice

Shadow AI grows wherever the sanctioned tools are painful. Report the friction; do not route around the control.

## 4. AI-enabled threats

Sub-area 7.4, AI-enabled threats.

**Deepfakes, cloned voices and model-written phishing. The old giveaways no longer work.**

### The first cloned-voice fraud

Reported by the Wall Street Journal, it is widely treated as the first known case of voice cloning used to commit fraud. The ear was fooled. Only procedure could have caught it.

**Asked:** A chief executive takes a call that sounds exactly like his German boss, and is told to pay a supplier urgently

**Answer:** In 2019, a UK energy firm transferred two hundred and twenty thousand euros before anyone realised the voice was synthetic

### The video call that was never real

Arup, a British engineering firm, later confirmed that fake voices and images were used. He had doubted the email that started it, but familiar faces on the call settled his fears. That is exactly what a deepfake is for.

**Asked:** A finance worker in Hong Kong joins a video call with people he believes are his company's senior staff

**Answer:** Every other person on that call was a deepfake; in 2024, he was talked into paying out around twenty-five million dollars

**You were taught to spot fakes by their flaws. A model removes the flaws, so the cues you learned are gone.**

### Phishing that reads perfectly

Business email compromise has cost organisations billions of dollars a year, according to the FBI's internet crime reporting. The craft that used to give a scam away is now the thing that makes it convincing.

**Asked:** An email arrives from a supplier, fluent and polite, referencing a real recent order

**Answer:** A model wrote it, tailored to you, and it is one of thousands sent in the same hour

Signs, and the checks that still work

| Threat         | Warning sign             | Your check                 |
|----------------|--------------------------|----------------------------|
| Deepfake video | Odd lighting or blinks   | Call the person back       |
| Cloned voice   | Urgency, strange request | Verify on a second channel |
| Phishing email | Familiar tone, new ask   | Check the real address     |
| Fake invoice   | Pressure to pay now      | Confirm with the supplier  |

## Stop and check

That combination is the signature of a social-engineering attack, AI-assisted or not. Slow down, verify out of band, and report it.

**You cannot out-spot a machine at faking. You can out-procedure an attacker, because procedure ignores how good the fake is.**

## Change of details

Any request to change payment details should be verified in person, or by a call to a number you already held. Never reply to the request itself.

## 5. Prompt injection

Sub-area 7.5, prompt injection.

**Prompt injection is text a model reads carrying its own hidden instructions, which the model then follows as if you had given them.**

## Not what you signed up for

The paper, by Kai Greshake and colleagues, is called 'Not What You've Signed Up For'. They demonstrated it against real, widely used systems. The model was not broken. It simply did what the text told it to do.

**Asked:** Researchers hid instructions inside ordinary web pages that an AI assistant had been asked to read

**Answer:** In 2023, they showed the assistant could be steered into following those hidden instructions instead of the user's own

Direct and indirect

| Type     | Where it comes from     | What happens                       |
|----------|-------------------------|------------------------------------|
| Direct   | The user                | Someone types the trick themselves |
| Indirect | Content the model reads | Hidden instructions are obeyed     |
| Both     | Any untrusted text      | The model cannot tell them apart   |

## The instruction in the fine print

Researchers have shown instructions can be hidden in white text, in comments, or in metadata. You see a document. The model sees a document full of orders.

**Asked:** You ask an assistant to summarise a long document somebody else sent you

**Answer:** Buried inside it, in text the same colour as the page, is an instruction aimed at the assistant

## **How a hidden instruction reaches you**

1. You ask the model to read a document or a page
2. That content contains hidden instructions
3. The model treats those instructions as a request
4. It acts on them, and you never see why

## **Never trust, always verify**

Treat anything a model produces after reading outside content as untrusted, especially if it suggests an action, a link, or a payment.

**Good systems keep your instructions and incoming content apart. Where they cannot, treat the output as untrusted.**

## **An assistant that acts**

An assistant that can send, book, buy or delete is worth far more to an attacker than one that can only talk.

**The model follows the text it is given. Prompt injection exploits that obedience, not a flaw in the model's intent.**

**The information is the risk. The tool only decides how far it travels.**

## **Domain 7 in six lines**

- Never enter protected, personal or operational data
- Assume inputs may be kept, learned from or reviewed
- Shadow AI leaks quietly; use the approved route
- Deepfakes and cloned voices are beaten out of band
- Prompt injection hides instructions inside content
- Verify anything a model says after outside reading

## Domain 8

# Ethics, Bias, Responsible AI & Governance

Weight 8% · 10 exam questions · about 18 minutes

Fairness, transparency, human oversight and the professional decision framework.

**AI does not make decisions. People do, with AI's help. Responsibility never moves to the machine.**

## 1. Bias and fairness

Sub-area 8.1, bias and fairness.

**A system learns from a world, and that world already contains our habits. Show it an unfair past and it will copy it forward.**

### The recruiting tool that taught itself

Reuters reported the story in 2018. Engineers could not fully fix it, and the project was scrapped. The system had learned exactly the world it was shown.

**Asked:** Amazon builds a system to review résumés and shortlist candidates for technical jobs

**Answer:** By 2015 it has taught itself to penalise the word “women's” — and downgrades two women's colleges

### The recidivism score

The overall accuracy was similar for both groups. The errors were not. That gap is the whole lesson about fairness.

**Asked:** A risk tool called COMPAS scores defendants in Broward County, Florida, for how likely they are to re-offend

**Answer:** ProPublica found the mistakes ran in opposite directions for black and white defendants

Four doors bias walks through

| Source   | What happens               | Example                   |
|----------|----------------------------|---------------------------|
| Data     | The past is copied forward | Old hiring patterns       |
| Coverage | Some groups are absent     | A rare dialect, unlearned |
| Design   | A proxy carries the bias   | Postcode for ethnicity    |
| Use      | Applied out of context     | Model moved to a new job  |

### The common trap

Accuracy asks whether it was right. Fairness asks who pays when it is wrong. A system can win on one and fail the other.

### Two ways to treat fairness

|  |                  |                    |
|--|------------------|--------------------|
|  | Fairness by hope | Fairness by design |
|--|------------------|--------------------|

|   |                                |                                |
|---|--------------------------------|--------------------------------|
| 1 | Assume accuracy is enough      | Test outcomes group by group   |
| 2 | Wait for a complaint to arrive | Set a standard before launch   |
| 3 | Assume the data is fine        | Measure again after deployment |
| 4 | Explain it away afterwards     | Fix it, and record the fix     |

## 2. Transparency, accountability and explainability

Sub-area 8.2, transparency, accountability and explainability.

**Transparency: you know AI is involved. Explainability: the reasoning can be described. Accountability: a person answers.**

Telling the three apart

| Duty           | The question it answers | The simple test     |
|----------------|-------------------------|---------------------|
| Transparency   | Was AI used here?       | Would they know?    |
| Explainability | Why this answer?        | Can you say why?    |
| Accountability | Who is responsible?     | Is a name attached? |

### The welfare system nobody could inspect

The court found the system was not sufficiently clear or verifiable, and that it risked discriminating. Opacity was the fatal flaw.

**Asked:** The Netherlands runs SyRI, an algorithm that combines government data to flag people for welfare-fraud checks

**Answer:** In February 2020, a court in The Hague halted it — partly because nobody could see how it worked

### The credit limit with no explanation

In March 2021 it found no unlawful discrimination — but criticised a lack of transparency that had shaken public trust. Being right was not enough.

**Asked:** Cardholders notice women receiving lower credit limits than their husbands, on otherwise similar finances

**Answer:** New York's financial regulator investigates, reviewing underwriting data for around 400,000 applicants

**Record what the system is, what it decided, who checked it, and where its limits lie. That record is your only honest answer.**

**Explainable does not mean publishing your code. It means you can give a straight account of why the system said what it said.**

## 3. Human oversight

Sub-area 8.3, human oversight.

**Someone must understand the output, have the power to reject it, and have the time to decide. A signature is not oversight.**

**Even the experts drifted**

The study was published in the journal Radiology in 2023. The pull of a confident machine is real, measurable, and it does not spare experts.

**Asked:** Researchers gave 27 radiologists mammograms to read, each with a purported AI suggestion beside it

**Answer:** When the AI suggested a wrong category, readers followed it — even the very experienced ones

## The common trap

Ask what would have to be true for this output to be wrong — then go and check that one thing. That is the antidote.

**A person who can change the outcome is in the loop. A person who can only watch is on the loop. Only one is oversight.**

## The right to a human

Regulators have been clear: you cannot satisfy the rule by “fabricating” human involvement. The oversight has to be meaningful.

**Asked:** A serious decision is made about a person by automated processing alone

**Answer:** Data protection law gives them the right to human intervention — and a token glance does not count

**If a person has no real power to overrule the system, they are not overseeing it. They are narrating it.**

## 4. Disclosure

Sub-area 8.4, disclosure.

**Whenever a person is affected, whenever AI drafted something you sign, whenever law or policy requires it, and whenever silence would mislead.**

**Withholding AI use removes the other person's ability to weigh what they are being given. The facts can be right and the impression still misleading.**

## The voice that was never there

There was no disclosure. There was a lie, dressed up to sound authoritative. The harm came directly from the absence of the truth.

**Asked:** In January 2024, robocalls used an AI clone of a president's voice to tell New Hampshire voters not to vote

**Answer:** The FCC ruled AI voices in robocalls illegal, and proposed a six million dollar fine against the consultant behind it

## The rule taking shape

Europe's AI Act requires that people are told when they are dealing with AI, and that deepfakes and AI-generated public-interest text are labelled as such.

## Good practice

One honest line in the covering note beats a paragraph of excuses later. “Drafted with an AI tool, checked by me” is usually enough.

**Say it where the reader will meet it — in the document, at the point of use. A footnote nobody reads is not disclosure.**

**Don't ask whether you can get away without saying anything. Ask whether the other person would want to know.**

## 5. Policy and governance

Sub-area 8.5, policy and governance.

**Policy says what is allowed. Governance proves it happened — with owners, records and review. Policy without governance is just a wish.**

Three layers of rule

| Layer     | What it does             | Example             |
|-----------|--------------------------|---------------------|
| Law       | Sets the hard boundaries | Data protection law |
| Policy    | Directs acceptable use   | Approved tools only |
| Procedure | Shows how to comply      | Record and review   |

### A framework for getting it right

NIST released the AI Risk Management Framework in January 2023. Crucially, the four functions form a loop, not a one-off checklist.

**Asked:** Organisations ask: how do we manage AI risk in a way that works, day after day?

**Answer:** The US standards body published a framework built on four functions: govern, map, measure, manage

**Somebody owns it, somebody records it, and somebody checks it — before deployment, and after.**

### The benefit system that went wrong

The Dutch data protection authority later fined the tax administration. This is what missing governance looks like at national scale.

**Asked:** Dutch authorities used risk-scoring to flag childcare-benefit fraud, and leaned in part on nationality as a signal

**Answer:** Thousands of innocent families were wrongly pursued; in 2021 the government resigned over the scandal

### The governance rhythm

1. Own: a named person for every system
2. Register: know what is in use, and where
3. Review: check before, and keep checking after
4. Respond: a route to challenge, and to fix

## 6. A professional decision framework

Sub-area 8.6, the professional decision framework.

**The framework is used before you accept the output. Used properly, it costs you a moment, not a procedure.**

### Six questions, in order

1. Purpose: is AI the right tool for this task at all?

2. Permission: does policy allow this use, with this tool?
3. Protection: is this data safe to put into the system?
4. Proof: have I checked the output against a source I trust?
5. Fairness: could this disadvantage a person or group unfairly?
6. Responsibility: can I explain and own this decision?

## The framework in action

Three of the six questions settled it, with no drama and no harm. That is the framework earning its keep — quietly, before anything went wrong.

**Asked:** A staff officer is asked to summarise a stack of reports for a briefing, and thinks of using a public AI tool

**Answer:** Purpose is fine; permission and protection are not. The officer stops, and uses the approved tool instead

## From can, to may, to should

- **What you can do** — What the tool is technically capable of
- **What you may do** — What policy and law permit
- **What you should do** — What fairness and your judgement demand

The framework, on one card

| Question       | Ask yourself                    | Red flag          |
|----------------|---------------------------------|-------------------|
| Purpose        | Is AI the right tool here?      | Used just because |
| Permission     | Does policy allow this use?     | Unapproved tool   |
| Protection     | Is this data safe to share?     | Sensitive data in |
| Proof          | Have I verified the output?     | Accepted unread   |
| Fairness       | Could this disadvantage anyone? | A group excluded  |
| Responsibility | Can I explain and own it?       | No name attached  |

## How it fits together

The six questions structure your thinking. They do not replace it. You still decide — and you still own the decision.

**Ask before you accept the output. If any answer is a red flag, stop and escalate. Record the answers for consequential work.**

**Bias is designed in or out. Humans stay accountable. Policy makes both routine — and six quiet questions keep you straight.**

## Domain 8 in seven lines

- Bias enters through data, coverage, design and use
- Accurate is not the same as fair
- Transparency, explainability and accountability are three duties
- A human owns every consequential decision
- Disclose AI use whenever people are affected

- Governance: a named owner, a register, and review
- Ask: purpose, permission, protection, proof, fairness, responsibility

## Glossary

| Term                                  | Meaning                                                                                                   |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Artificial intelligence</b>        | Machines performing tasks we associate with human intelligence; in practice, learning patterns from data. |
| <b>Machine learning</b>               | The part of AI that improves from data rather than following rules a person wrote.                        |
| <b>Deep learning</b>                  | Machine learning using layered neural networks.                                                           |
| <b>Generative AI</b>                  | Deep learning that produces new content: text, images, audio or code.                                     |
| <b>Narrow AI</b>                      | A system built for one task or a narrow family of tasks. All deployed AI today is narrow.                 |
| <b>General AI</b>                     | A hypothetical system matching a person across any task. It does not exist today.                         |
| <b>Token</b>                          | A chunk of text, roughly part of a word, that a language model processes.                                 |
| <b>Context window</b>                 | The amount of text a model can consider at once.                                                          |
| <b>Knowledge cut-off</b>              | The date after which a model has no built-in knowledge.                                                   |
| <b>Embedding</b>                      | A numeric representation of meaning that lets a model compare items by similarity.                        |
| <b>Hallucination</b>                  | Fluent, confident output that is simply wrong.                                                            |
| <b>Prompt</b>                         | The instruction you give a model.                                                                         |
| <b>Few-shot prompting</b>             | Including examples in a prompt to show the model the pattern you want.                                    |
| <b>Retrieval-augmented generation</b> | Supplying a model with retrieved documents so its answer is grounded in real sources.                     |
| <b>Agent</b>                          | A system that plans and takes multiple steps towards a goal using tools.                                  |
| <b>Shadow AI</b>                      | Using unapproved AI tools for official work.                                                              |
| <b>Prompt injection</b>               | Hidden instructions inside content that trick a model into ignoring its real task.                        |
| <b>Bias</b>                           | A systematic unfairness in a model's output, usually inherited from its data or design.                   |
| <b>Human oversight</b>                | A person remaining responsible for consequential decisions.                                               |
| <b>Disclosure</b>                     | Telling people when AI has been used to produce something that affects them.                              |

How to use this handbook — read the domain section before its video, then again afterwards, and use the key-point recaps for rapid revision. The videos and this handbook teach the same material in the same order.